

STOCKHOLMN SKAVSTA, Sweden

WMO: 024853

Lat: 58.789N

Lon: 16.912E

Elev: 140

StdP: 14.62

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.1	8.7	-0.5	5.4	3.9	5.1	7.2	10.1	28.4	40.9	24.8	39.5	4.9	320	0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	80.5	65.5	77.1	64.2	73.8	62.5	68.2	75.6	66.2	73.6	64.5	72.0	10.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	96.3	71.4	64.0	89.7	69.7	61.2	81.0	67.6	32.4	76.0	30.9	72.8	29.6	72.2	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.6	19.8	17.6	DB	-3.8	84.7	7.6	3.4	-9.3	87.1	-13.8	89.1	-18.0	91.0	-23.5	93.5	
			WB	-4.3	70.5	7.6	2.4	-9.8	72.2	-14.2	73.7	-18.4	75.0	-24.0	76.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.4	27.8	28.4	33.4	41.8	50.6	58.4	63.5	62.1	54.1	44.5	36.6	30.6
	DBStd	14.37	8.94	8.62	7.48	6.25	6.37	4.95	4.89	4.96	5.61	6.31	6.87	9.05
	HDD50	3353	689	603	515	257	71	2	0	0	25	188	403	600
	HDD65	7587	1154	1023	979	697	447	203	89	116	327	634	853	1065
	CDD50	1316	0	0	0	9	89	255	417	377	149	19	1	0
	CDD65	76	0	0	0	0	0	7	41	27	1	0	0	0
	CDH74	687	0	0	0	1	19	98	373	190	6	0	0	0
	CDH80	95	0	0	0	0	1	16	58	20	0	0	0	0
Wind	WSAvg	8.3	9.0	9.0	8.8	8.3	8.3	8.2	7.6	7.3	7.9	8.4	8.4	8.8
Precipitation	PrecAvg	21.0	1.6	1.0	0.9	1.2	1.5	1.4	2.6	2.6	1.9	1.9	2.5	2.0
	PrecMax	32.0	3.0	2.2	2.0	2.4	2.8	3.0	4.7	8.5	4.1	5.5	5.1	5.6
	PrecMin	14.8	0.3	0.3	0.0	0.2	0.3	0.2	0.7	0.6	0.8	0.5	0.9	0.3
	PrecStd	4.3	0.7	0.6	0.6	0.6	0.9	0.7	1.2	2.0	0.9	1.3	1.3	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.6	48.0	57.5	68.2	76.8	82.2	84.5	82.6	75.0	61.1	53.9	51.5
		MCWB	44.6	43.8	47.0	54.2	58.8	63.8	66.6	67.0	63.2	56.0	50.4	47.8
	2%	DB	43.0	44.5	52.0	62.5	71.5	76.7	81.0	78.9	70.0	58.7	50.4	46.5
		MCWB	40.9	41.4	44.1	50.8	57.3	62.1	65.9	65.8	60.9	54.4	48.1	44.5
	5%	DB	40.9	41.3	48.0	57.6	67.7	73.3	78.6	75.5	66.5	55.8	48.3	44.4
		MCWB	39.1	38.8	41.9	48.0	56.0	60.1	64.8	64.7	58.9	52.5	46.9	42.6
	10%	DB	38.9	39.2	44.3	53.7	63.0	69.9	75.1	72.0	64.1	53.8	46.3	41.3
		MCWB	37.5	37.3	39.6	45.4	53.1	58.9	63.7	63.2	57.7	50.8	45.2	40.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.8	44.5	48.2	55.5	62.0	66.5	70.8	71.3	65.3	57.6	51.9	48.4
		MCDB	45.9	46.1	56.1	66.2	71.0	76.1	78.0	77.7	71.0	60.2	53.9	50.8
	2%	WB	41.3	41.4	44.6	51.5	59.0	63.8	68.5	68.4	62.0	55.3	48.8	44.6
		MCDB	42.3	43.0	50.4	60.2	67.5	73.0	76.5	75.0	67.9	57.8	50.2	45.6
	5%	WB	38.9	39.3	42.1	49.1	56.5	61.6	66.8	66.2	60.1	53.6	46.7	42.2
		MCDB	39.9	40.8	46.6	57.1	65.5	70.2	74.8	72.4	65.2	56.2	48.0	43.2
	10%	WB	37.0	37.1	39.7	46.3	54.4	60.0	65.0	64.5	58.5	51.8	44.8	40.1
		MCDB	38.1	38.6	42.8	51.9	62.3	67.8	73.1	70.9	63.0	54.0	45.5	41.0
Mean Daily Temperature Range	5% DB	MDBR	9.0	10.7	13.9	17.7	18.9	18.0	18.0	17.6	16.0	12.1	8.6	8.8
		MCDBR	9.6	11.6	19.3	25.3	25.5	23.8	23.5	21.6	19.6	13.7	11.6	9.4
		MCWBR	8.9	9.3	12.8	15.4	13.8	12.2	10.5	10.9	11.8	10.4	10.4	8.8
	5% WB	MCDBR	8.9	10.2	17.7	23.1	22.3	20.5	19.4	18.8	17.5	12.3	10.3	8.8
		MCWBR	8.5	8.8	12.4	14.5	12.9	11.8	10.1	10.7	11.1	10.0	9.5	8.5
Clear-Sky Solar Irradiance	taub	0.284	0.321	0.338	0.375	0.372	0.361	0.389	0.380	0.347	0.337	0.310	0.270	
	taud	2.245	2.305	2.364	2.325	2.404	2.462	2.405	2.426	2.498	2.426	2.283	2.172	
	Ebn at Noon	171	222	255	263	273	278	266	259	252	215	158	132	
	Edh at Noon	14	22	28	34	33	32	33	30	24	20	14	11	
All-Sky Solar Radiation	RadAvg	104	293	769	1259	1665	1841	1685	1340	892	399	134	71	
	RadStd	14	49	81	123	136	89	155	124	104	61	29	9	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.92	N/A	N/A	N/A	N/A	N/A	-290	-329	N/A	N/A
Regional (53 neighbors)		+0.68	N/A	N/A	N/A	N/A	N/A	-220	-268	N/A	N/A

Nomenclature: See separate page